Prediction of Chronic Disease Risk Using Nutrition, Lifestyle & Clinical Data from The Irish Longitudinal Study on Ageing (TILDA)

Abderahman Haouit

CSIS Research Centre

Department of Computer Science and Information Systems

Faculty of Science and Engineering

University of Limerick

Submitted to the University of Limerick for the degree of

Artificial Intelligence (MSc) 2026

1. Supervisor: Dr. Alison O'Connor
Computer Science & Information Systems
University *of* Limerick
Ireland

Abstract

Background: Chronic diseases such as Cardiovascular Disease (CVD), Diabetes Mellitus (DM), Frailty (FR), and obesity represent major contributors to morbidity in older adults. While traditional prediction models focus heavily on clinical and functional indicators, the contribution of nutritional and lifestyle factors to early risk detection remains underexplored.

Methods: This thesis applies machine learning techniques to data from The Irish Longitudinal Study on Ageing (TILDA), a national representative longitudinal study of older adults. Five waves of socio demographic, clinical, nutritional, psychological, and behavioural data, with selected variables from the externally produced Research and Development Corporation (RAND) + Harmonised TILDA dataset, together were systematically cleaned and standardised using a multi stage preprocessing pipeline. Predictive models, including (*models x, y, z*) were developed to estimate the risk of chronic disease and FR related outcomes. Model performance was evaluated using (*mention examples*), and feature importance was used to assess predictive value.

Results: TBD

Conclusions: TBD

Keywords: ageing, (Artificial Intelligence (AI)), Machine Learning (ML), predictive modelling, Frailty (FR), chronic disease, nutrition, The Irish Longitudinal Study on Ageing (TILDA)

Declaration

I Abderahman Haouit declare that I have produced this thesis without the prohibited assistance of third parties and without making use of aids other than those specified; notions taken over directly or indirectly from other sources have been identified as such. This thesis has not previously been presented in identical or similar form to any other Irish or foreign examination board.

The thesis work was conducted from 2025 to 2026 under the supervision of Dr. Alison O'Connor at University of Limerick.

Limerick, 2026

Acknowledgements

I would like to acknowledge my supervisor, Dr. Alison O'Connor, for her guidance, support, and for providing a well organised environment throughout the implementation of this project.

I would also like to thank the UL Science and Engineering Research Ethics Committee for approving the ethical aspects of this research. Special thanks to the Irish Social Science Data Archive (ISSDA) team for facilitating access to the TILDA dataset, collected by Trinity College Dublin.

Your dedication goes here. Open the dedication.tex file found at:
0_frontmatter/dedication.tex

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List of Abbreviations

AI	Artificial Intelligence
AUC	Area Under the ROC Curve
AUROC	Area Under the Receiver Operating Characteristic Curve
BMI	Body Mass Index
BP	Blood Pressure from Sphygmomanometer
brainPAD	Brain predicted age difference
CAPI	Computer Assisted Personal Interview
CES-D	Center for Epidemiologic Studies Depression Scale
COG	Scales and Questions on Cognition
CRT	Choice Reaction Time (Cognitive Test)
CV	Cross Validation
CVD	Cardiovascular Disease
DIS	Disability, Functional Impairment and Helpers
DM	Diabetes Mellitus
DNN	Deep Neural Network
EU	European Union
F1	F1-score (harmonic mean of precision and recall)
FOF	Fear of Falling
FR	Frailty
GBM	Gradient Boosting Machine
GDPR	General Data Protection Regulation
HADS	Hospital Anxiety and Depression Scale
HGS	Hand Grip Strength

IADL	Instrumental Activities of Daily Living
INC	Income Variables
IPAQ	International Physical Activity Questionnaire
ISSDA	Irish Social Science Data Archive
JSON	JavaScript Object Notation
MD	Medication Data
MGS	Maximum Gait Speed
MH	Scales and Questions on Mental Health and Wellbeing
ML	Machine Learning
MVPA	Moderate-to-Vigorous Physical Activity
NOS	Newcastle-Ottawa Scale
OR	Odds Ratio
PA	Physical Activity
PMF	Pseudonymised Microdata File
RAND	Research and Development Corporation
Recall	Recall (true positive rate)
RF	Random Forest
SCQ	Self Completion Questionnaire
SCR	Screening Tests
SHAP	Shapley Additive Explanations
SLR	Systematic Literature Review
SOC	Social Variables
SPSS	Statistical Package for the Social Sciences
SR	Systematic Review
TILDA	The Irish Longitudinal Study on Ageing
TUG	Timed Up and Go (mobility test)
UGS	Usual Gait Speed
WHO	World Health Organization

List of Abbreviations

Chapter 1

Introduction

Ageing populations experience substantial changes across physical, nutritional, cognitive, and psychosocial domains, each shaping vulnerability to chronic disease and functional decline [6–8]. Understanding how these multidomain factors interact over time is essential for early risk identification.

The financial and societal burden is substantial. In Europe, poor diet is a major risk factor for non communicable diseases, contributing billions in healthcare costs annually [9]. In Ireland, malnutrition alone is estimated to cost over €1.4 billion per year [10]. In response to these challenges, international public health frameworks, including the World Health Organization (WHO) *Global Action Plan* on noncommunicable diseases and Ireland's *Healthy Ireland* framework, highlight nutrition and lifestyle factors as central components of disease prevention strategies.

European regulatory frameworks such as the General Data Protection Regulation (GDPR) and proposed European Union (EU) Artificial Intelligence (AI) Act emphasise transparency and highlight the need for interpretable Machine Learning (ML) models in healthcare.

Traditional risk prediction models have largely relied on clinical biomarkers such as Blood Pressure from Sphygmomanometer (BP) and lipid profiles. Yet, growing evidence shows that broader predictors, including nutrition [11], Physical Activity (PA) [12], Scales and Questions on Mental Health and Wellbeing (MH) [13], and Social Variables (SOC) [14] improve predictive performance. Estimating the likelihood of developing a condition given certain risk factors helps quantify the influence of these predictors in clinical decision making.

TILDA[1] provides a national representative cohort of adults aged 50+ in Ireland, surveyed repeatedly across multiple domains. For cross study comparability and methodological consistency in model development, selected variables from the externally produced RAND Harmonised TILDA dataset (available for Waves 1 and 2) were included.

The harmonised dataset provides standardised variable definitions and documented construction rules, which are particularly relevant in a machine learning setting where models are trained and evaluated on partitioned data. Its inclusion supports robust feature engineering, reproducible train validation splits, and alignment with prior ageing related ML studies that adopt harmonised frameworks. The specific role of harmonised variables within the modelling pipeline is detailed in Chapter 3.

Despite these advantages, existing ML research on chronic disease prediction faces persistent limitations: over reliance on clinical predictors, limited incorporation of lifestyle and SOC variables, predominantly cross sectional study designs that fail to capture trajectories, and a lack of interpretability consistent with emerging AI governance standards.

This thesis addresses these gaps by leveraging longitudinal TILDA data across multiple domains, using interpretable ML methods to combine modifiable health behaviours, psychosocial factors, and established clinical measures to enhance prediction of ageing related outcomes.

1.1 Research problem and motivation

Despite increasing research and clinical interest in lifestyle, nutrition factors and advances in ML methods, approaches to chronic disease prediction in older adults remain limited in several ways.

[1]TILDA's dataset are pseudonymised and accessed under license via ISSDA. Use adheres to UL S&E ethics approval and data retention requirements.

- **Limited Integration of Multidomain Predictors:** Existing models focus primarily on clinical measurements and often neglect modifiable behavioural, dietary, and psychosocial factors [11, 15].

- **Underuse of Longitudinal Dynamics:** Few studies leverage repeated measures to capture developments of risk, such as FR progression [16]. Probabilistic reasoning can help estimate the likelihood of developing a condition at a future time given earlier measurements, for example, calculating the probability $P(\text{FR at wave 5} \mid \text{UGS at wave 1}, \text{Nutrition at wave 2})$.

  This approach helps quantify how early indicators influence later outcomes and supports better feature selection and timing of interventions in predictive models.

- **Lack of Explainability:**Many ML models are not interpretable, limiting clinical adoption and failing to meet emerging regulatory requirements for trustworthiness AI in healthcare [17].

Addressing these challenges is essential to improve predictive accuracy, support evidence based interventions, and ensure alignment with evolving policy and regulatory priorities.

1.2 Research aims and objectives

The goal of this thesis is to develop and validate ML models that integrate nutrition, lifestyle, and MH features with clinical data to improve prediction of chronic disease and ageing related outcomes in a representative cohort of older adults. This goal will be achieved through the following objectives:

- Review existing applications in ML for chronic disease prediction in ageing populations, with emphasis on gaps in lifestyle and nutrition integration.

- Develop a reproducible pipeline to preprocess and align TILDA waves 1–5, and engineer multidomain features (nutrition, activity, mobility, MH, clinical indicators).

- Develop and compare baseline statistical models (e.g., logistic regression) with advanced interpretable ML methods such as Random Forest (RF) and Gradient Boosting Machine (GBM).
- Evaluate models using predictive classification performance metrics including Area Under the ROC Curve (AUC), calibration and error measures (accuracy, precision, Recall (true positive rate) (Recall), F1-score (harmonic mean of precision and recall) (F1)), perform ablation analyses to assess the incremental value of lifestyle and nutrition predictors.
- Identify actionable predictors (e.g., vitamin D status, UGS reserve) and highlight modifiable risk factors for intervention.
- Document ethical, governance, and reproducibility practices for handling longitudinal pseudonymised health data.

1.3 Contributions of the research

This research makes contributions at three levels:

- **Empirical:** Demonstrates how behavioural, dietary, and other MH features complement established clinical measures to enhance predictive performance.
- **Methodological:** Provides a reproducible pipeline for aligning longitudinal TILDA data across domains and integrating diverse features into ML workflows.
- **Clinical:** Highlights actionable modifiable factors alongside traditional clinical indicators for targeted interventions and policy.

1.4 Thesis Outline

In *Chapter* 2 a literature review on ML prediction of chronic disease in ageing populations, with emphasis on lifestyle, nutrition predictors and methodological considerations, is presented. *Chapter* 3 discusses the TILDA study design, data preprocessing, alignment, and feature engineering. In *Chapter* 4 the modelling framework is outlined. This includes baseline and advanced ML, hyperparameter tuning, validation, and evaluation metrics. *Chapter 5* shows the model result, including model comparisons, assessments of feature importance, and an ablation analysis. In *Chapter 6* the implications, limitations, and ethical considerations are discussed, and future directions for the research are proposed. Additional supporting documents are included in appendix A.

Chapter 2

Literature Review

This chapter reviews literature on ageing, chronic disease, and predictive modelling using TILDA (introduced in Chapter 1). It focuses on three interrelated domains central to later life health: FR, physical function, MH, and nutrition within the context of biological ageing. This review focuses on FR, MH, and nutrition as key determinants of later life outcomes. In line with the WHO healthy ageing framework, research using TILDA highlights FR, MH, and nutrition as interconnected determinants of later life outcomes. The following subsections examine each domain in turn.

2.1 Methodology

Several sources describe how systematic and integrative reviews are commonly carried out in fields such as software engineering, health, and education. The approach followed key principles for systematic mapping, for example, defining research questions, identifying relevant studies through structured database searches, applying inclusion and exclusion criteria, and categorising results as outlined in [1]. The integrative literature review [18] helped structure and connect findings across studies, while [19, 20] guided the organisation of ideas. In related methodological work, [2] clarified how SRs can identify research gaps, and [21] showed how mapping and synthesis highlight trends and future directions. This combined guidance informed a structured yet flexible methodology suitable for ageing in AI, and health research.

2.1.1 Search strategy and database screening

The main topic of interest was chronic disease prediction using nutrition, lifestyle, and clinical data from TILDA. Relevant databases (e.g., ScienceDirect, PubMed) were searched using combinations of keywords such as "Irish Longitudinal Study on Ageing", "machine learning," "chronic disease", "nutrition", "diet", and "physical activity" (see table 2.1).

The search covered studies published from 2011 to 2025. Figure 2.1 summarises the number of studies identified per year, highlighting trends and gaps over time.

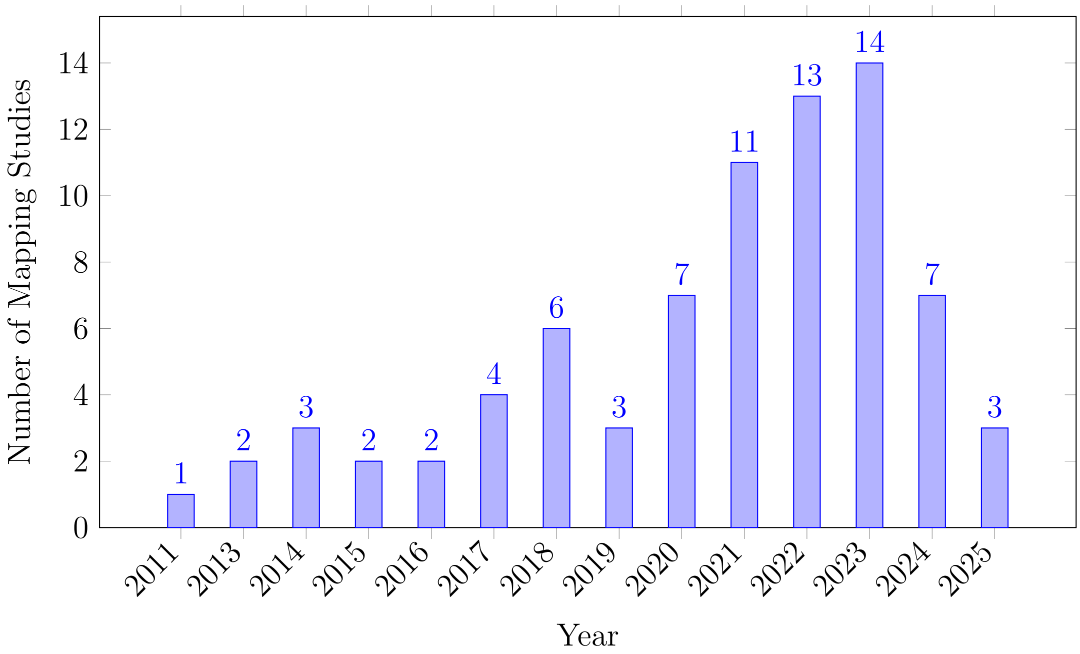

Figure 2.1: Publications per year.

Some search strings returned few or irrelevant results, particularly for ML related TILDA studies, highlighting a key research gap. Iterations continued until suitable combinations consistently returned relevant studies. Duplicates were removed using Zotero's reference management feature. The remaining studies were individually and manually checked for relevance to "ageing," "chronic disease," and ML. Using Zotero's notes functionality, the initially created table 2.3 captured key elements for each paper, including model type, features used, and

Table 2.1: Database searches and number of studies per database 2.3 [1].

Database	**Search**	**Results**
Oxford Academic	(TILDA OR Irish Longitudinal Study on Ageing)AND (chronic disease OR diabetes OR cardiovascular) AND (nutrition OR diet OR "physical activity")	1081
	(TILDA OR "Irish Longitudinal Study on Ageing") AND ("machine learning" OR "predictive model" OR "risk prediction")	71
IEEE Xplore	TILDA OR "Irish Longitudinal Study on Ageing"	56
	(TILDA OR "Irish Longitudinal Study on Ageing") AND ("machine learning" OR "predictive model" OR "risk prediction")	7
ScienceDirect	(TILDA OR "Irish Longitudinal Study on Ageing") AND ("chronic disease" OR cardiovascular OR diabetes) AND (nutrition OR diet OR lifestyle OR "physical activity")	257
	(TILDA OR "Irish Longitudinal Study on Ageing") AND ("machine learning" OR "predictive model") AND ("chronic disease" OR diabetes) AND (nutrition OR "physical activity")	20
PubMed	(TILDA OR "Irish Longitudinal Study on Ageing") AND ("chronic disease" OR cardiovascular OR diabetes) AND (nutrition OR diet OR lifestyle OR "physical activity")	27
	(TILDA OR "Irish Longitudinal Study on Ageing") AND ("machine learning" OR "predictive model" OR "risk prediction")	9

main findings for each paper in the collection, as illustrated in the PRISMA flow diagram (Figure 2.2).

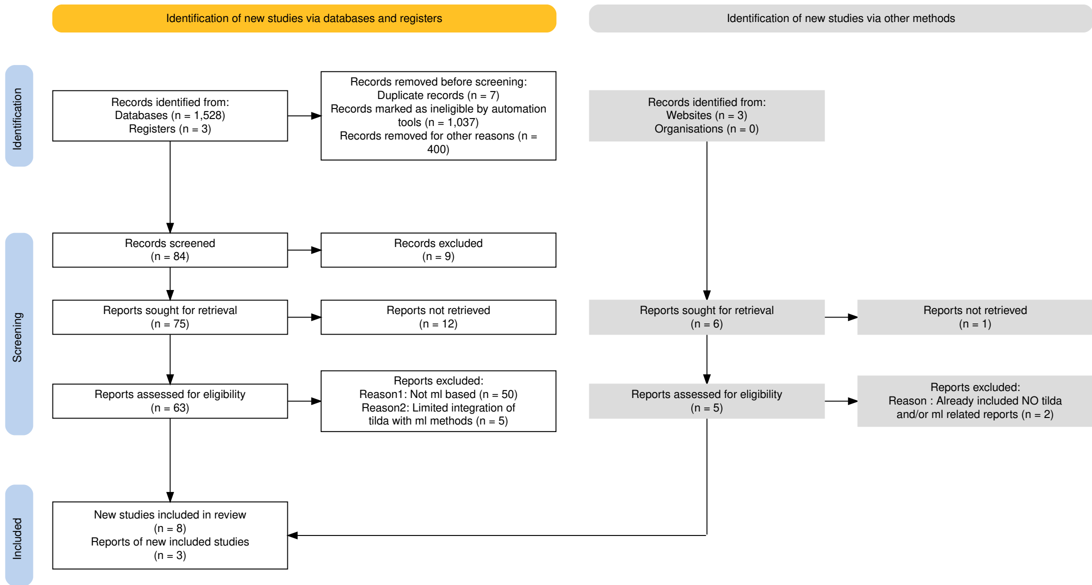

Figure 2.2: PRISMA chart for SRs showing the study selection process for the systematic mapping and integrative review [5].

2.1.2 Quality and relevance evaluation

To systematically assess the quality of the included studies, a customisable version Newcastle-Ottawa Scale (NOS) was applied; see table 2.2. The NOS evaluation studies across three domains: *Selection* (S1–S4), *Comparability* (C1–C2), and *Outcome* (O1–O3). *Selection* criteria assess the representativeness of the sample, clarity of exposure measurement, and temporal alignment of outcomes. *Comparability* examines the control of key confounders (variables that may influence both the predictor and outcome), while *outcome* evaluates measurement validity, follow up adequacy, and completeness. Each domain is scored 0–1, recall the Bernoulli random variable (0 = criterion not met, 1 = criterion met), and the total score reflects the overall study quality.

This initial table guided a further refinement of the results into two sub tables: (1) studies critically related to TILDA, (2) studies using TILDA data ML for predictive contexts, which revealed minimal to no integration with TILDA.

Table 2.2: Quality evaluation of SLRs [2].

Reference	S1	S2	S3	S4	C1	C2	O1	O2	O3	Total
(Xu, 2023 [22])	1	1	1	1	1	1	1	1	1	9
(Donoghue, 2023 [17])	1	1	1	1	1	1	1	0	1	8
(Moguilner, 2021 [16])	1	1	1	1	1	0	1	1	1	8
(Davis, 2021 [6])	1	1	1	1	1	0	1	1	0	7
(Boyle, 2021 [23])	1	1	1	1	1	0	1	0	1	7
(Tzemah-Shahar, 2022 [24])	1	1	1	0	1	0	1	1	1	7
(A. Shirsath, 2024 [25])	1	1	1	1	1	1	1	0	1	8
(Romero-Ortuno, 2021 [26])	1	1	1	1	1	1	1	1	1	9
(Bardon, 2020 [27])	1	1	1	0	1	0	1	1	1	7
(Zhang, 2025 [28])	1	1	1	1	1	1	1	0	1	8
(Sutin, 2023 [29])	1	1	1	1	0	0	1	1	1	7

Table 2.3 summarises selected studies on ageing using TILDA and ML, reflecting key gaps and opportunities identified during systematic mapping. These steps were aligned with the research matrix created at the beginning, ensuring that the refinement of studies and synthesis of findings stayed closely tied to the original research goals.

2.1.3 Synthesis and Integration

Using the NOS scores, each study was annotated in Zotero with details on dataset, key variables, analytical approach, and main findings, facilitating a consistent and systematic evaluation across studies. For the systematic component, data was systematically extracted and synthesised to identify trends (e.g., Markov models, logistic regression, 5-fold Cross Validation (CV)) and research gaps [2], complemented by DoctoralWriting resources [30, 31], which clarified how to define research questions and develop logical arguments.

The integration of a research matrix with systematic mapping and integrative synthesis ensured that the methodology was well suited to capturing the complexity of ageing, nutrition, chronic disease risk and predictive modelling studies focused primarily on TILDA based studies, but due to the limited number of ML

Table 2.3: Summary of selected studies on ageing using TILDA.

Theme	Author/year	Dataset	Variables/features	Methods/model type	Key Findings/outcome	Limitations
FR	Romero-Ortuno (2021) [26]	Wave 1–5	FR state, FR phenotype (FP) components (Exhaustion(EX), Unexplained weight loss(WL), Weakness(WK), Slowness(SL), Low physical activity(LA))	Multi state Markov model, longitudinal 8 year follow up	32% of pre frail to non frail; FR strongly predicted 31% mortality	Self report bias; observational design
Falls + ML	Donoghue (2023) [17]	Wave 1–3	Gait, mobility, medications, history of falls	Conditional inference forest; 5-fold CV; undersampling for imbalance	Accuracy up to 69.7%; for adults aged 50–64.	Area Under the Receiver Operating Characteristic Curve (AUROC) values were <= 0.69 and PRAUC <= 0.39 indicating low predictive accuracy; no external validation; cross sectional; Class imbalance
Fear of Falling (FOF)	Zarudsky (2014) [32]	Wave 1	Age, sex, fall history	Cross sectional prevalence analysis	23.3% prevalence; higher in women; 70% with FOF had no recent falls in the last 12 months	Self reported; cross sectional
Functional Mobility	Huang (2019) [33]	Wave 1–2	Measured by Timed Up and Go (mobility test) (TUG), Hand Grip Strength (HGS), physical activity, vision, Body Mass Index (BMI), age	Linear regression, stratified analyses	Age modifies effects; PA and grip protect mobility; vision benefit only in middle aged (Beta = -0.032, P = 0.002)	Observational; 2 year follow up
Depression	Jacob (2023) [34]	Wave 1–2	Falls, FOF, HADS-A, Center for Epidemiologic Studies Depression Scale (CES-D)	Prospective logistic regression	Falls predicted incident anxiety Odds Ratio (OR)=1.58 and depression OR=1.43	Self reported falls; short follow up; mediation effect partially explored
	Laird (2023) [15]	Wave 1–5	Moderate-to-Vigorous Physical Activity (MVPA), walking, sociodemographics	Longitudinal regression; mixed model	Higher physical activity reduced depressive symptoms; dose response effect	Self reported PA; observational; limited causal inference
Nutrition	Murphy (2023) [11]	Wave 1–5	Plasma lutein, zeaxanthin	Linear/ordinal logistic regression	Higher carotenoids linked to slower FR progression Lower odds of becoming frail (ORs = 0.57–0.89)	Observational; not predictive of longitudinal muscle changes
	Bardon (2020) [27]	Wave 1–2	Hospitalisation, functional status, SOC support, sex, falls	Logistic regression; sex stratified analyses	Functional decline, hospitalisation, poor support increased malnutrition risk; sex differences	Observational(short follow up (2 years)); limited generalisability ¿65 years, self reported
ML	Xu (2023) [22]	Wave 1–3	105 predictors from 40,000+ variables (SOC, clinical, mental, family)	Stacked ensemble: RF, XRT, GLM, GBM, Deep Neural Network (DNN); Shapley Additive Explanations (SHAP) explainability	AUC=0.854 for 2 year DM prediction; top predictors LDL, cirrhosis, sex	No external validation; short follow up; ensemble complexity limits deployment
	Davis (2021) [6]	Wave 3	Age, HGS, chair stands, BMI, cognitive tests	Histogram gradient boosting regression; SHAP	r^2 test: UGS≈0.41, Maximum Gait Speed (MGS)≈0.46, GSR≈0.21	Modest predictive power; cross sectional; no external validation
	Zhang (2025) [28]	Longitudinal IMAGEN cohort	Psychopathology, personality, cognition, substance use, environment	Regularised logistic regression; diagnostic	Longitudinal AUC: ED=0.71, MDD=0.64, AUD=0.67; shared transdiagnostic predictors	Limited age range; no external replication
	Boyle (2021) [23]	Wave 3	Brain predicted age difference (brainPAD); processing speed; attention; cognitive flexibility; verbal fluency	Linear regression, correlation analyses	Higher brainPAD associated with lower cognitive performance across multiple domains	Cross sectional; no causal inference; limited sample

applications within TILDA, additional studies on older adult populations and chronic disease were included.

2.2 Frailty and physical function

FR reflects declining physiological reserve, providing a framework for understanding physical vulnerability in ageing. In TILDA, it is captured through both the FR phenotype and the FR index, with UGS, HGS, and mobility decline serving as practical predictors. These measures strongly predict hospitalisation and mortality.

2.2.1 Probabilistic interpretation of frailty

One can argue that from a probabilistic perspective, the chance of an adverse outcome (such as mortality, Y) given observed features (such as FR markers, X) is a conditional probability $P(Y \mid X)$. This allows us to deduce which FR features are most predictive of adverse outcomes. Features may be independent, dependent, or mutually exclusive.

The presence or absence of metabolic syndrome represents a mutually exclusive event that can inform risk models.

$$P(X_i \cap X_j) = P(X_i)P(X_j)$$

for independent features X_i and X_j, guiding feature selection and reducing redundancy in model inputs. This reflects the likelihood that an individual with certain FR characteristics will experience a particular outcome.

Evidence suggests that FR is dynamic, not irreversible. For example, multi state modelling showed that many prefrail adults transition back to non frail, although established FR strongly predicts mortality [26]. Metabolic syndrome accelerates the development of FR, suggesting a metabolic pathway of vulnerability [35]. The presence of metabolic syndrome modifies the conditional probability $P(\text{FR} \mid \text{Metabolic Syndrome})$ and may increase the likelihood of adverse outcomes over time. Another detail is that age matters: obesity predicts mobility decline earlier in life, whereas later in life, traits that protect against this, including HGS, become more significant [33].

These findings partly conflict: FR appears both reversible and progressive, depending on risk context. This complexity highlights the need for probabilistic models that can update beliefs about risk as new features are observed, as formalised by Bayes theorem:

$$P(Y \mid X) = \frac{P(X \mid Y)P(Y)}{P(X)}$$

where $P(Y \mid X)$ is the posterior probability of outcome Y given features X.

2.2.2 Bayesian updating of frailty risk

Causal inference is limited by observational designs, and predictors are frequently examined separately, ignoring interactions. Most FR models have only been with the TILDA dataset, and with this study that focuses on the same TILDA but uses both longitudinal TILDA waves and harmonised variables to improve cross study comparability, enabling for a enabling for a more robust training validation splits in ML models, and explores richer predictive patterns over time. These gaps point to the need for multidimensional approaches. Traditional regression models may not fully capture age dependent interactions, suggesting a role for more flexible methods such as machine learning techniques and models.

2.3 Mental health in older adults

MH, including depression, anxiety, loneliness, and SOC isolation, strongly shapes quality of life. Evidence from TILDA shows strong links to both physical and SOC factors. [36] found that participation in physical, SOC, and cognitive activities summarised by the Act-Belong-Commit (ABC) indicator was associated with lower depression and anxiety. This suggests SOC integration and lifestyle protect mental wellbeing. However, [37] reported that self reported measures may exaggerate such associations..., raising concerns of bias. These two findings conflict: the effect of SOC engagement is strong when self reported and weaker when objectively assessed.

Objective data strengthen the case for activity. Accelerometer based measures show a dose response link between moderate to vigorous activity and lower depressive symptoms [12]. Yet reverse pathways are also evident: incident falls and functional decline predict later depression, refereed by FOF [34]. Together, these results reveal a bidirectional relationship, MH both influences and is influenced by physical decline. Most studies rely on broad self report scales, such as CES-D or Hospital Anxiety and Depression Scale (HADS), with little biomarker or clinical validation.

This literature suggests MH in ageing cannot be modelled as a one way effect. It requires longitudinal, multidimensional approaches that can capture mutual pathways, an area where machine learning is well suited, especially when combined with probabilistic models that estimate the likelihood and posterior probability of MH outcomes given multiple interacting features.

For example, we can estimate:

$$P(\text{Depression} \mid \text{Activity, Falls, SOC Engagement}),$$

quantifying how these interacting features together influence MH outcomes. Probabilistic modelling helps determine which features have the strongest effect and which may be redundant for predictive purposes.

2.4 Nutrition and biological ageing

Nutrition interacts with both biological and psychosocial systems, shaping ageing trajectories and FR. Biomarkers like vitamin D, lutein, and zeaxanthin have been linked to physical and cognitive outcomes in TILDA. [11] showed that higher plasma lutein and zeaxanthin dietary carotenoids found in leafy vegetables were linked to slower FR progression. By contrast, [38] found vitamin D deficiency predicted DM, musculoskeletal decline, especially among adults with low sunlight exposure. These studies point to different nutritional mechanisms: antioxidant protection versus metabolic regulation.

2.4.1 Bayesian perspective on nutritional biomarkers

The effect of nutrition on ageing outcomes can be formalised probabilistically using Bayes theorem:

$$P(\text{Outcome} \mid \text{Nutritional Status}) = \frac{P(\text{Nutritional Status} \mid \text{Outcome})P(\text{Outcome})}{P(\text{Nutritional Status})},$$

where the likelihood function $P(\text{Nutritional Status} \mid \text{Outcome})$ quantifies how observed nutritional biomarkers relate to health states, and the posterior $P(\text{Outcome} \mid \text{Nutritional Status})$ captures updated beliefs about outcomes, including uncertainty in small sample studies. This helps identify which biomarkers are most informative for predicting ageing outcomes which supports feature selection in ML models.

2.4.2 Contextual and integrative considerations

Malnutrition also reflects SOC and functional risks. [27] found that hospitalisation, functional loss, and poor SOC support increased malnutrition risk, with notable sex differences. This highlights a limitation in much nutritional research: biological markers are studied without integrating SOC context.

The difference between biological and chronological age is a major point of debate. Biomarker based metrics that better capture these cumulative exposures, like FR indices or epigenetic clocks, frequently vary from chronological age. McCarthy and Azizi link metabolic syndrome and nutritional deficits to accelerated biological ageing [39, 40]. Although the majority of the evidence is cross sectional and has little ability to capture changes over time, these indicators offer a more precise prediction of vulnerability. Reliability is also decreased in biomarker research with small sample sizes, which can be formalised through the likelihood function $P(\text{Data} \mid \theta)$ and its impact on posterior uncertainty in Bayesian inference.

Taken together, nutrition interacts with biological, metabolic, and SOC systems. Isolating single nutrients risks oversimplification. Integrative modelling that combines biomarkers, clinical factors, and SOC determinants supported by machine learning pipelines offers a more promising route. Models that estimate the joint distribution of features and outcomes can better capture the complex dependencies relevant to biological ageing.

2.5 Machine Learning in ageing and chronic disease

ML is increasingly applied in ageing research because it can capture non linear and multidimensional relationships that traditional regression models, such as logistic regression, often miss [22, 23].

2.5.1 Feature sets in ML studies

Studies using ML in ageing research include many types of predictors: clinical measures, sociodemographic factors, lifestyle habits, cognitive tests, and psychosocial variables. A recurring pattern, however, is that models often underline easily measured physical health data, such as comorbidity counts, mobility scores, or vital signs, while more complex but important predictors, including nutrition and SOC factors, are underrepresented [32]. This imbalance may limit the ability of models to capture the full complexity of chronic disease risk with ageing, especially considering that MH and nutrition exert independent and interacting effects on later life outcomes (here). Models that focus only on ready available clinical data risk reproducing the same biases seen in traditional regression approaches.

2.5.2 ML models used

The methodological approaches used in ageing research span both traditional and advanced models. Logistic regression and Markov models remain common baselines due to their interpretability and simplicity, yet they are often unable to capture non linearities or higher order interactions. More recent work has shifted towards ensemble methods such as RF, gradient boosting machines GBM, as well as DNNs, which generally achieve higher predictive accuracy.

Similarly, [41] evaluated mortality prediction using a standard logistic regression model with demographic, clinical, and functional predictors, achieving an

AUC of 0.78, whereas ensemble methods in [22] reached 0.854 for DM prediction, showing how flexible, non linear models can better capture complex interactions among ageing related predictors.

2.5.3 Performance and Validation

Predictive performance across studies is generally moderate to strong, with AUC values between 0.70 and 0.90 for outcomes like DM, UGS, depression, and mortality. This indicates that models can distinguish reasonably well between individuals who will or will not experience an outcome.

However, there are important limitations. Many studies rely only on internal validation, testing models on the same data used for training. This can inflate performance estimates and reduce confidence in applying the model to new populations. To add, most studies are cross sectional, so causal or temporal effects cannot be assessed. When combined with the underuse of nutrition and SOC predictors, these issues indicate that current models may not fully represent the multidimensional nature of ageing and could produce context specific results rather than generalisable predictions [32].

2.5.4 Explainability in ML for ageing

As ML models become more complex, interpretability emerges as a critical challenge. Explainability tools like SHAP can highlight feature importance, but their use is inconsistent, particularly in deep learning models [4, 6, 22]. These methods improve transparency and can highlight unexpected drivers of outcomes, such as psychosocial variables being stronger predictors than biomedical ones, yet black box concerns remain, especially for deep learning models, where feature interactions are difficult to track. For ageing research, where clinical adoption requires both accuracy and trust, explainability is not optional but essential. Current efforts show promise but remain fragmented, with no consensus on best practice. This gap links directly to the broader methodological limits discussed in the next section on gaps in the literature.

2.6 Gaps in the Literature

First, most studies rely on TILDA or similar cohorts of Irish and predominantly older adults `(50+)`. This limits confidence in how well results transfer to other populations. This thesis focuses on TILDA studying ageing in depth but also makes use of the harmonised dataset, which links TILDA with comparable UK and US cohorts. This helps place findings in a broader international context while retaining a core emphasis on older adults.

Second, there is an over reliance on physical health predictors such as comorbidity counts and mobility measures, while nutrition, SOC, religious or spiritual, and environmental variables are often overlooked. As shown in [11], nutritional biomarkers like lutein and zeaxanthin are strong predictors of FR, yet these are rarely integrated into broader ML pipelines. By contrast, studies such as [27] highlight the role of SOC support and functional decline in malnutrition risk. Together, these examples show that excluding nutrition and SOC factors risks producing models that reinforce the same biomedical biases seen in regression heavy work. From a Bayesian viewpoint, the omission of key features reduces the explanatory power of the model and weakens the posterior probability $P(\text{Outcome} \mid \text{All Features})$ compared to $P(\text{Outcome} \mid \text{Limited Features})$. Including a richer set of features increases the likelihood that the model accurately reflects the true data generating process.

Third, methodological choices remain conservative. Regression models remain dominant, likely due to their interpretability and ease of implementation. However, they often fail to capture non linear interactions and higher order dependencies present in ageing related data. Comparisons with more advanced approaches such as DNNs or transformers are rarely explored. For instance, [41] achieved an AUC of 0.78 using logistic regression for mortality prediction in TILDA, whereas [22] reported an AUC of 0.854 with ensemble methods for DM incidence. This contrast illustrates that more flexible ML models can provide superior predictive performance while maintaining interpretability when combined with explainability techniques. Where else ensembles or brainPAD approaches have been attempted, validation is almost always internal, with little cross cohort testing. Weak validation and limited feature sets reduce the generalisability of the estimated conditional probabilities and increase the risk of overfitting. Maximum likelihood estimation and Bayesian inference can provide a formal framework for assessing model fit and updating predictions as more diverse and representative data become available.

Fourth, many outcomes relevant to older adults such as pain, sleep quality, and subjective wellbeing remain underexplored. For example, [11] focused on plasma biomarkers for FR without assessing sleep or pain, and UGS models such as [6] similarly omit subjective quality of life. Omitting these outcomes limits the real world application of predictive models. Including broader outcomes in ML pipelines can better capture holistic ageing trajectories and improve interpretability for interventions targeting older adults.

Fifth, cross sectional bias and data handling weaken reliability. Much nutritional work, such as [11], relies on single time point biomarker data, which cannot capture longitudinal change. By contrast, studies applying multi wave TILDA data [26] demonstrate the added value of modelling temporal dynamics. Similarly, missing data are often handled through listwise deletion, as in [41], which risks biasing results toward healthier participants, whereas more robust imputation approaches remain underused. Missing or poorly handled data can distort the likelihood function and hence the posterior predictions, reducing confidence in the model's outputs.

Finally, interpretability challenges persist. While tools like SHAP have been applied to highlight feature importance [4, 22], their use is inconsistent, and consensus on best practice has yet to emerge. Without transparent explanations, even accurate models may face barriers to clinical acceptance. Taken together, these examples show that ageing research continues to rely on narrow predictors, conservative models, and weak validation. Addressing these gaps requires integrative, diverse, and transparent approaches that link multidimensional predictors with explainable ML.

2.7 Chapter Summary

This chapter reviewed evidence on ageing and chronic disease across three domains, FR, physical function, MH, nutrition, and biological ageing. We also examined how machine learning has been applied to predict later life outcomes. The findings showed that while FR is partly reversible, it remains strongly tied to mortality, that MH both shapes and is shaped by physical decline, and that nutrition interacts with biological, metabolic, and SOC systems in ways that are often underestimated.

While machine learning has improved prediction accuracy in ageing research, challenges remain in integrating diverse predictors and ensuring interpretability and longitudinal validation.

Addressing these challenges requires the integrative, longitudinal, and transparent approaches that combine diverse predictors. These gaps motivated the feature engineering and modelling strategies implemented in Chapters 3 and 4, including the use of harmonised variables to ensure reproducibility and generalisability, for ML modelling.

Chapter 3

Technical Approach

3.1 Study Design

TILDA provides longitudinal data across five waves (Waves 1–5) for community dwelling adults in Ireland. For contextual comparison and benchmarking against international cohorts see Chapter 1. An external, harmonised TILDA dataset, was also inspected and prepared. The harmonised dataset, which has been pre aligned to international standards e.g., the Gateway to Global Ageing Data (G2G), contains a subset of variables mapped to a common cross study format and thus differs from the raw TILDA files. Although the harmonised dataset was simpler to process, it was still subjected to the variable filtering and numerical standardisation steps developed as part of this research to ensure compatibility with the five wave cleaned datasets prior to modelling.

3.2 Cohort Definition

All participants who provided data in at least one TILDA wave were considered eligible. To maximise the available sample for machine learning, no exclusions were applied on the basis of age, medical conditions, mobility, cognitive status, or health centre attendance.

Participants were removed only in the rare case that, following manual variable filtering, no analytically usable variables remained for that individual. Filtering occurred primarily at the *variable level* during the manual screening stage, which involved features removed for extreme missingness, lack of variation, or non informative structure. This approach preserves the largest possible sample while ensuring all retained inputs are analytically meaningful for downstream analysis and model development.

- **Participant inclusion:** All respondents who contributed data in at least one wave were included in the initial sample.
- **Usable data requirement:** Participants were excluded only if all retained variables were missing for that individual after manual variable screening.
- **Longitudinal retention:** Individuals were retained regardless of how many waves they contributed to, enabling initial maximum dataset utilisation.
- **No clinical exclusions:** No participant was removed due to medical diagnoses, cognitive scores, functional limitations, or non attendance at the health centre assessment.

This approach prioritises data integrity and sample size for machine learning applications, differing from exclusion-heavy cohort definitions often used in traditional epidemiological studies [42].

3.3 TILDA Data Collection and Screening

TILDA data collection occurred over five longitudinal waves as detailed in Table 3.1.

Each wave included three primary components:

1. **Computer Assisted Personal Interview (CAPI):** Home based, computer assisted personal interviews covering health, socio demographics, and family.
2. **Self Completion Questionnaire (SCQ):** Paper based self completion questionnaires capturing sensitive information on mental health, lifestyle, and wellbeing.

3. **Health Assessment:** Clinical and biomarker measurements conducted in dedicated centres or at home (Wave 1 and Wave 3).

All TILDA data were pseudonymised prior to release, meaning that personal identifiers such as names and addresses were removed or replaced with unique codes to protect participant confidentiality. Each participant received a unique identifier ('id'), along with household and cluster identifiers ('household', 'cluster'), allowing longitudinal linkage across waves without revealing individual identities.

Pseudonymisation ensures that analyses comply with data protection standards while retaining the structural and analytical integrity of the dataset. From a data cleaning and machine learning perspective, pseudonymised identifiers enable accurate merging of data across waves, preserve the longitudinal design, and support reproducible preprocessing and feature engineering. While these identifiers cannot be used for analyses of personal demographics like names or addresses, this limitation does not affect the modelling of health outcomes, lifestyle factors, or other variables of interest.

Prior to any data cleaning or modelling, a structured multi stage workflow was implemented to screen and reduce the dimensionality of the raw TILDA datasets. This workflow separates automated diagnostic profiling from manual variable exclusion and ensures transparency in all variable selection decisions.

The process consisted of the following steps:

1. **Raw data loading:** Raw TILDA PMF files were loaded without cleaning or recoding. Waves 1–5 were loaded from Statistical Package for the Social Sciences (SPSS) (`.sav`) files, while the harmonised dataset was loaded from a (`Stata`) (`.dta`) file due to technical compatibility issues.

2. **Automated variable profiling (Python):** For every variable, summary statistics were computed programmatically, including data type, proportion of missing values, number of unique values, most frequent value, and frequency of the most frequent value. These summaries were exported to spreadsheet format for inspection.

3. **Manual variable screening (spreadsheet based):** Using predefined rule based thresholds, variables were manually reviewed and flagged for exclusion based on missingness, lack of variation, dominance of a single category, or administrative function.

4. **Creation of retained variable lists:** Variables not flagged for exclusion were programmatically concatenated in `.xlsx` and manually transferred into wave specific JavaScript Object Notation (JSON) files representing the retained `variables of interest`.

5. **Reloading filtered datasets:** The JSON files were then used to subset the original raw datasets, producing reduced datasets with identical rows but fewer variables. No cleaning, recoding, or imputation was applied at this stage.

These reduced datasets form the direct input to the automated preprocessing and modelling pipeline described in Chapter 4.

Table 3.1: Dimensions of raw TILDA and harmonised datasets (prior to filtering).

Dataset	Collection Period	Participants (N)	Raw Variables (N)
Wave 1	2009–2011	8501	1619
Wave 2	2012–2013	7206	724
Wave 3	2014–2015	6397	1029
Wave 4	2016	5713	990
Wave 5	2018–2019	4978	983
Harmonised	2016 (release)	8501	572

Following the five step workflow outlined above, Python scripts were used to profile all variables across each wave. For every variable, the following metrics were computed: data type, number of missing values, number of unique values, most frequent value, and frequency of the most frequent value. These summaries were exported to spreadsheet format to support structured manual review prior to variable dropping.

Manual filtering was applied according to the following criteria:

- **Missingness:** Drop variables with > 80% missing values. This threshold balances retaining sufficient feature coverage with ensuring statistical reliability. Variables exceeding this threshold provide limited information in most cases, increase imputation uncertainty, and reduce effective sample size, which is particularly critical in longitudinal machine learning analyses.

- **Unique values:** Drop variables with a single observed value, as they contain no discriminative information and cannot contribute to modelling or inference. Variables with low cardinality (2–4 unique values) were retained for review, as they frequently represent meaningful categorical responses (e.g., binary indicators or ordinal scales) relevant to ageing outcomes.

- **Dominant category (top frequency):** Variables where a single category (sample value) occurs in over > 90% of observations were flagged. Such dominance indicates limited variability, reducing analytical utility and predictive value. Variables exceeding this threshold were marked for review or exclusion.

- **Identifiers / administrative fields:** IDs, version tags, and sampling weights were identified using the PMF metadata. Variable names and descriptions were exported from each wave's PMF file into CSV files (code provided in Appendix A) to systematically document and review features prior to filtering. These exports enabled structured manual inspection and traceability between raw PMF metadata and downstream feature selection decisions.

As part of the manual screening and filtering workflow. Variables exceeding exclusion thresholds (e.g., high missingness, single value, dominant category, administrative identifiers) were visually flagged (e.g., marked in red). Non flagged variables were programmatically concatenated using spreadsheet formulas (e.g., `=TEXTJOIN(", ", TRUE, J2:Jn)`) and then manually transferred into wave specific JSON files for each wave. These files represent the variables of interest retained after manual screening and filtering and serve as inputs for the upcoming automated data preprocessing & cleaning in Chapter 4.

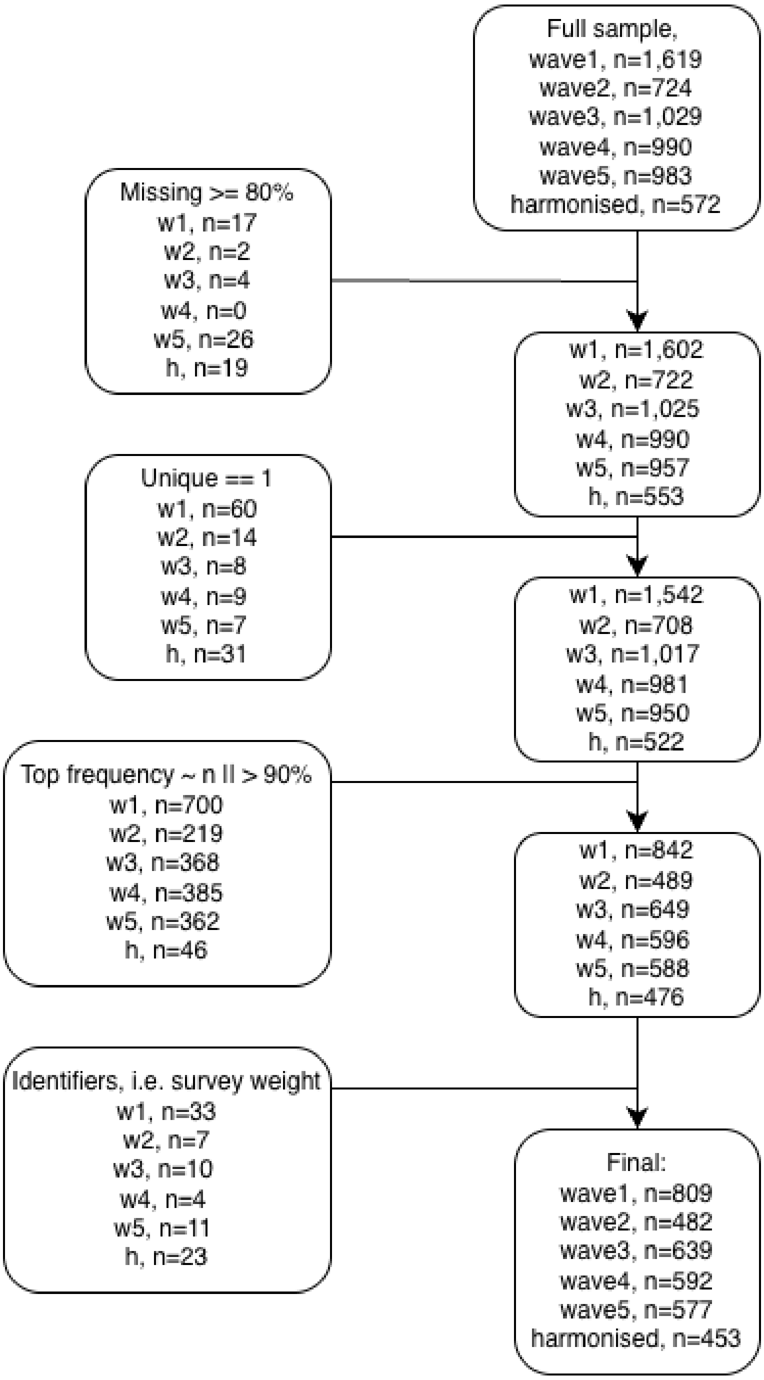

Figure 3.1: Flowchart of the five step workflow for variable screening and filtering. Placeholder counts indicate the progression of remaining variables per dataset from raw to filtered datasets after each step, ready for automated preprocessing.

The retained variables, defined in JSON files, were subsequently used to subset the raw datasets. This produced reduced CSV datasets per wave, containing only the variables of interest. At this stage, no cleaning or recoding was performed. The operation served purely as dimensionality reduction, preparing the data for the automated nine-scenario cleaning pipeline described in Chapter 4.

Importantly, all variable exclusion decisions were completed at this stage. Subsequent preprocessing steps focus exclusively on numerical standardisation, recoding, and preparation for machine learning rather than further feature removal. With the variables of interest selected and the datasets reduced to a manageable dimensionality, the next stage focuses on automated preprocessing, recoding, and preparation for machine learning.

3.4 Thematic Grouping

After manual filtering, the retained variables were organised into six thematic domains based on ageing literature and study objectives. Each variable's prefix, as defined in the PMF metadata (Table 3.2), indicates its source or research domain. This systematic assignment ensures that each thematic group contains conceptually related features and provides a clear structure for downstream automated preprocessing and machine learning.

Each variable in TILDA was assigned a prefix indicating its data source or research domain. Home based CAPI variables use a two letter module code followed by a three digit question code (e.g., `ph121`). Variables with multiple responses or loops include an underscore and number to specify the response or iteration (e.g., `ph201_05`).

Variables from the self completion questionnaire are prefixed with SCQ, while derived variables constructed from either CAPI or SCQ data use uppercase codes corresponding to their research area (Table 3.2). These prefixes provide a consistent framework for categorising features and guiding the assignment into thematic domains for downstream analysis.

Using the PMF prefixes as a guide, variables were assigned to six thematic groups (Table 3.3), each capturing a conceptually coherent domain: physical,

Table 3.2: PMF Derived variable prefixes [3].

Variable Prefix	**Research Area**
ADL	Activities of daily living
BEH	Health behaviours
BLOODS	Results of blood sample analysis
BP	Blood pressure from sphygmomanometer
COG	Scales and questions on cognition
CRT	Choice reaction time (cognitive test)
DIS	Disability, functional impairment, helpers
FR	Frailty, gait, exercise, falls and fractures
G2G	Gateway to Global Aging Data
GRIP	Grip strength
IADL	Instrumental activities of daily living
INC	Income variables
MD	Medication data
MH	Scales and questions on mental health and wellbeing
SCR	Screening tests
SES	Social class and socio
SOC	Social

cognitive, social, and behavioural aspects of ageing. This structure supports reproducible preprocessing and helps machine learning models focus on meaningful, longitudinally robust features.

Specifically, the thematic grouping:

1. Provides a clear framework for automated cleaning and filtering rules.
2. Ensures that model input variables are conceptually aligned with study objectives.

The six thematic groups integrate information from CAPI, SCQ, and derived variables, covering socio demographic context, physical health, functional status, lifestyle behaviours, nutrition/metabolic biomarkers, and psychological/cognitive measures. This structured grouping enhances interpretability, ensures comprehensive domain coverage, and provides a foundation for downstream predictive modelling and risk assessment.

Table 3.3: Thematic grouping of longitudinal features.

ID	Thematic Group	Description	Representative Variables
1	Socio Demographic & Context	Baseline Stratification Factors	Age
			Gender
			Education
			Income Variables (INC)
			Marital Status
			SOC
2	Physical Health & Clinical Baseline	Self reported and clinical indicators	Self Rated Health
			Screening Tests (SCR)
			BMI
			BP
			Comorbidity Count Medication Data (MD)
3	Functional Status & FR	Measures of mobility and physical decline	FR Index
			UGS
			Falls History
			Instrumental Activities of Daily Living (IADL)
			Disability, Functional Impairment and Helpers (DIS)
4	Lifestyle & Behavioural	Modifiable health related behaviours	Smoking
			Alcohol Intake
			PA
			International Physical Activity Questionnaire (IPAQ)
5	Nutrition & Metabolic Biomarkers	Objective nutritional and metabolic markers	Plasma Carotenoids
			Lutein
			Zeaxanthin
			Vitamin D/B12
			Cholesterol
6	Psychological & Cognitive	Mental health and cognitive performance	CES-D
			Loneliness
			Scales and Questions on Cognition (COG)
			Choice Reaction Time (Cognitive Test) (CRT)

Chapter 4

Implementation Details

4.1 Data Preprocessing

The preprocessing pipeline's primary aim was to convert all features, regardless of their original datatype (float, categorical, object, or mixed formats), into a standardised numerical representation suitable for machine learning. All nonresponse and sentinel values were consistently mapped to the numerical placeholder `-1.0`, and all valid entries were converted to `float64`. This ensured compatibility across the TILDA waves and enabled uniform downstream operations for feature engineering and modeling.

4.1.1 Input data shape (post manual filtering)

The dimensions of the manually filtered datasets, which serve as the input for the automated cleaning engine, are in table 4.1. This initial manual filtering step removed noninformative features (e.g., columns with zero variation, unique participant IDs, and nonessential administrative fields), resulting in a substantial reduction of up to half `50%` of the original raw variable count (e.g., from 1619 to 809 in Wave 1). While the Harmonised dataset uses a reduced subset of variables pre aligned to G2G standards, it was subjected to the same deterministic cleaning steps to ensure numerical consistency with the full five wave datasets used for modelling.

Although many Harmonised TILDA variables appeared superficially numeric, inspection of the raw values revealed that they frequently consisted of prefixed numeric labels followed by descriptive text (e.g., "1.Rarely or none of the time (less than 1 day)", "4.All of the time (5–7 days)"). While these formats might suggest that simple string stripping could recover numeric values, such an approach would have introduced semantic errors for censored range, calendar, frequency, and other hybrid response formats. For example, a Harmonised value such as "5–7 days" would reduce to `5` if stripped, whereas the appropriate numerical representation for consistency with the five wave cleaning logic is the midpoint of the range.

Applying the full pipeline guaranteed numerical consistency and prevented subtle distortions that would otherwise arise from inconsistent handling of hybrid string numeric representations.

Table 4.1: Input shape for cleaning pipeline (post manual variable filtering).

Dataset	Row (participants)	Column (features)
Wave 1	8501	809
Wave 2	7206	482
Wave 3	6397	639
Wave 4	5713	592
Wave 5	4978	577
Harmonised	8501	453

4.2 Cleaning Engine

The core challenge in converting heterogeneous variable types across five (including harmonised) waves into a single numerical standard required constructing a reproducible pipeline to achieve *numerical homogeneity,* referring to the property that all retained features are represented as consistent numerical values (float64) with a unified missing/sentinel code (`-1.0`), enabling uniform downstream processing. This was accomplished by developing a two stage system: *variable sce-*

nario mapping, specifically inside the "detect_column_scenario_mapping" function, C and the final *six step deterministic cleaning engine* D.

4.2.1 Stage 1: Variable scenario mapping and validation

An exploratory visualiser was used to systematically inspect the raw data distributions, unique values, and data type inconsistencies across all waves. This audit initially identified between five and seven distinct data scenarios, which were expanded and refined during development to nine total scenarios to cover the full scope of data heterogeneity: from simple numeric fields and numeric data with symbols to Likert type scales, calendar based frequency strings, and complex mixed string numeric fields. These nine initial scenarios were then systematically collapsed and categorised into the six final cleaning steps, reflecting an ordering based on transformation confidence.

The pipeline's first stage utilised an automated function (see appendix C), powered by the `re` (Regex) Python library, to classify every raw variable into one of these nine scenarios. During development, a dedicated "unknown complex" category was crucial: features routed here indicated a failure of the detection logic. This iterative debugging process, which identified and fixed classification errors, significantly drove the robustness of the final detection mechanism by ensuring the pipeline correctly classified features that initially appeared ambiguous.

To ensure robustness, the classification output was immediately processed by a validation routine that generated a comprehensive dictionary of all pre and post transformed features, along with their corresponding sample values. This dictionary was organised by the detected scenario after each cleaning step facilitated for manual inspection. The dictionary generated inside the "show_grouped_vars_for_scenarios" function was formatted to create an automatic friendly ".md" file for rapid, surface level verification, which honed the final mapping dictionary D. This process of using automated detection followed by visual, manual validation was essential for correcting errors in classification logic before the subsequent scenario transformations were executed in the pipeline.

4.2.2 Stage 2: Deterministic six step cleaning sequence

The nine input scenarios were processed by the *six step cleaning engine*. Resolving simple and unambiguous cases first increased the proportion of clean numerical features before addressing complex structures, which involved the censored and multi-class scenarios (figure 4.1). This deterministic sequence applied the same logic across all waves to ensure consistent data alignment (see appendix D).

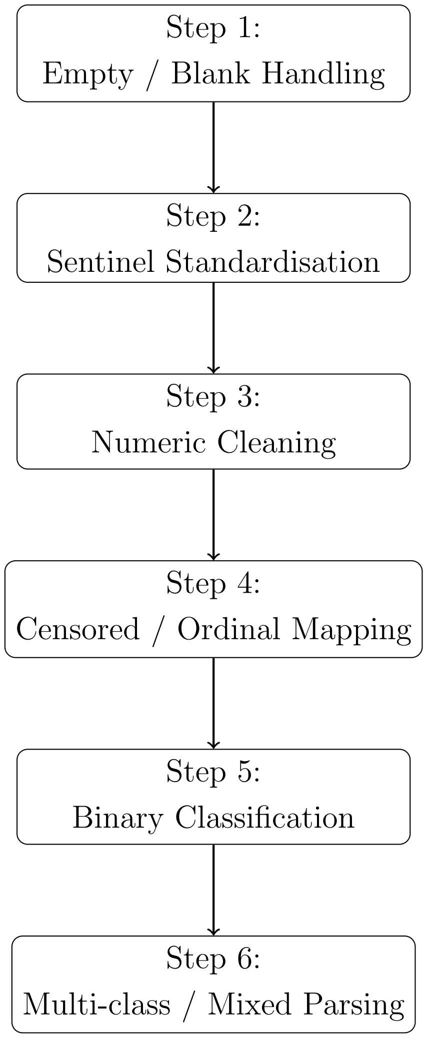

Figure 4.1: Flowchart of the final six step cleaning pipeline applied to all TILDA waves. See full transformation tables in appendix B.7 and D for detailed scenario progression code at each step.

A key difficulty in cleaning the five TILDA waves was the lack of guaranteed ordering in categorical values for the same variable across different waves (e.g., `Wave 1 feature(x)`: [`‘Very Good’, ‘Fair’, ‘Excellent’, ‘Good’, ‘Poor’, ‘DK’, ‘RF’`] and `Wave 3 feature(x)`: [`‘Excellent’, ‘Fair’, ‘Very Good’, ‘RF’, ‘Poor’, ‘DK’, ‘Good’`]). If the cleaning engine relied on a positional index (like traditional label encoding) for transformation, this inconsistency would result in different numerical codes for the same semantic category across waves (e.g., `Wave 1` $\rightarrow$ [`4.0, 2.0, 5.0, 3.0, 1.0, -1.0`], `Wave 3` $\rightarrow$ [`4.0, 2.0, 5.0, 3.0, 1.0, -1.0`]). To overcome this, the cleaning engine was designed to be *order invariant* wherever possible, particularly in the later steps. For multi class scenarios (Step 6), this required an explicit, wave independent lookup based on the category string itself to ensure that a category string, once transformed to a numerical value, remained consistent regardless of its appearance order in the raw data.

4.2.3 Scenario steps and their technical focus

1. **Empty / Blank Values:** Converts all `NaN`, empty strings, and whitespace to the standard non response value `-1.0`.

2. **Sentinel Values:** Standardises wave specific non response codes (e.g., `-98`, `-99`) and documented textual sentinels (e.g., `"dk"`, `"not applicable"`) from the TILDA PMF release guide to the unified `-1.0` placeholder.

3. **Numeric Transformation:** Removes non numeric characters (units, symbols, white spaces) and converts censored or annotated numeric strings (e.g., `"50+"`, `"<10"`, `"1-2.5"`) to a numerical float equivalent (e.g., boundary value or midpoint).

4. **Censored Categorical:** Applies explicit, known ordinal transformations and predefined mappings to features where text labels represent a latent numerical or ordered scale (e.g., health rating scales, Likert ranges). This step was intentionally performed manually for a subset of low count features to

build intuition and reliable mapping dictionaries for the more complex subsequent scenarios. This included conversions for calendar based frequency strings like `"twice a week"` to a numerical weekly equivalent.

5. **Binary Categorical:** Automatically maps two level features (e.g., Yes/No, Male/Female) to `1.0/0.0`. Importantly, this mapping is *order invariant*: the numerical assignment does not depend on the original ordering of category labels in the raw data (e.g., `[Male, Female]` versus `[Female, Male]`). Instead, the encoding relies on *semantic detection rules* (e.g., assigning `0.0` to negated or absence indicating strings such as `"no"`, `"not"`, or wave specific refusal codes, under context constraints) alongside *count based detection rules* (where a variable exhibits exactly two non missing unique values). This design choice ensures consistent binary representation across waves and features, independent of label ordering or survey specific value presentation.

6. **Multi-class / Mixed (The Complex Scenario):** This final step processes all remaining multi label, mixed type, and string numeric hybrid columns. The challenge was made more complex by the fact that the order of categorical string values could vary across waves, as discussed previously. To handle this, numerical assignments were made using an explicit dictionary lookup based on the original order of the sample values , rather than relying on positional or alphabetical order. This ensured consistent numerical representation. The step performs extensive lexical processing, dedicated number extraction, and rule based mapping to convert dense, unstandardised text into numerical floats, ensuring all residual variable types are successfully quantified. The transformed values were subject to increased manual verification due to the variable complexity.

4.3 Cleaning Validation

The robustness of the automated cleaning pipeline was evaluated in two stages. First, the internal consistency of cleaned predictors was assessed against established biological relationships. Second, full model replication was performed using a secondary study to confirm predictive utility.

4.3.1 Pairwise correlation validation

Correlation analysis was conducted on the seven primary predictors identified by [4] as explaining 95% of the peak (GSR) model performance, which is defined in this study as the difference between maximum and usual gait speed. Since the GSR outcome itself is not natively available in the public Wave 3 dataset, validation was focused on the directional relationships between the predictors themselves. This internal consistency check ensures that the biological and clinical signals reported in the literature are preserved within the dataset (post cleaning).

Figure 4.2 displays the correlation matrix for the selected predictors. These top 7 predictors were used to establish directional relationships for verifying that the numerical parsing pipeline specifically the handling of sentinel values was successful.

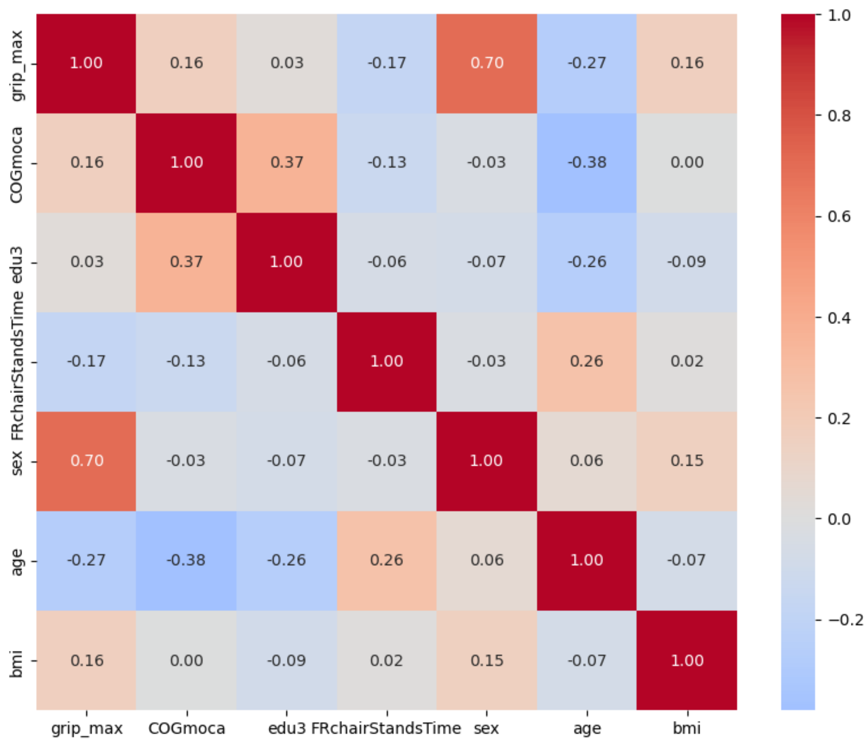

Figure 4.2: Correlation heatmap for top 7 predictors (Wave 3).

Table 4.2: Directional validation of the seven features explaining 95% of GSR variance as identified by [4].

Predictor 1	**Predictor 2**	**Expected Direction / Justification**	**Observed** r
Grip Strength (GRIP)	Age	**Negative (-)**; GRIP is a positive predictor of GSR, while age is a negative predictor [4].	-0.265
GRIP	Chair Stands Time	**Negative (-)**; Higher strength (grip) is expected to correlate with faster (lower) performance time in physical tests [4].	-0.170
Montreal Cognitive Assessment (MOCA) Score	Age	**Negative (-)**; Cognitive performance (MOCA) contributes positively to GSR, whereas increasing age is associated with reduced reserve [4].	-0.381
Education	MOCA Score	**Positive (+)**; Third level education and cognitive scores are both cited as positive contributors to cognitive reserve and GSR [4].	0.367
BMI	Age	**Weak / Mixed**; BMI and age both negatively impact GSR, though their internal correlation in older cohorts is typically weak [4].	-0.073

4.3.2 Model Validation

Full model validation was performed by replicating the methodology of [16], which predicts 8–year mortality across Waves 1–5 using baseline Wave 1 predictors. The replication of AUC scores in the range of `0.70-0.90` serves as a functional validation.

Chapter 5

Empirical Evaluation

You can also you this kind of referencing if needed in the text:

Chapter 6

Conclusion

6.1 Summary

6.2 Conclusions

6.2.1 Data availability Statement

The data analysed in this study is subject to the following licenses/ restrictions: The datasets generated during and/or analysed during the current study are not publicly available due to data protection regulations but are accessible at TILDA on reasonable request. Requests to access these datasets should be directed to https://tilda.tcd.ie/data/accessing-data/.

6.2.2 Ethics Statement

The studies involving human participants were reviewed and approved by Faculty of Health Sciences Research Ethics Committee at Trinity College Dublin, Ireland. The patients/ participants provided their written informed consent to participate in this study.

6.3 Contributions

6.4 Future Work

Appendix A

Python code for exporting PMF metadata to CSV

```
import pandas as pd
import pyreadstat
from pathlib import Path
import json

data_files = {
    "wave1.json": "data/0053-01_TILDA_Wave1_2009-2011/0053-01_TILDA_Wave1_Data/SPSS/0053-01_TILDA_Wave1_PMF_v1.12.sav",
    "wave2.json": "data/0053-01_TILDA_Wave2_2012-2013/Data/0053-02_TILDA_Wave2_PMF_v2.7.sav",
    "wave3.json": "data/0053-01_TILDA_Wave3_2014-2015/Data/0053-04_TILDA_Wave3_PMF_v3.6.sav",
    "wave4.json": "data/0053-01_TILDA_Wave4_2016/Data/0053-05_TILDA_Wave4_PMF_v4.4.sav",
    "wave5.json": "data/0053-01_TILDA_Wave5_2018/Data/0053-06_TILDA_Wave5_PMF_v5.5.sav",
    "harmonized.json": "data/0053-06_TILDA_Harmonized_2016/Data/0053-03_TILDA_Harmonized_v1.2.dta",
}


def load_df_with_var_of_interest(
    var_dir="var_of_interest", data_files=data_files, verbose=False
```

```python
):
    dfs = {}
    var_dir = Path(var_dir)

    for json_name, data_path in data_files.items():
        file = var_dir / json_name
        with open(file, "r") as f:
            vars_list = json.load(f).get("variables_of_interest")
            if verbose:
                print(f"Variables of interest from {file.name}: {vars_list[:10]} ...")

        if data_path.endswith(".sav"):
            df_full, meta = pyreadstat.read_sav(
                data_path, encoding="latin1", apply_value_formats=True
            )
            df, _ = pyreadstat.read_sav(
                data_path,
                usecols=vars_list,
                encoding="latin1",
                apply_value_formats=True,
            )
        else:
            df_full = pd.read_stata(data_path, convert_categoricals=True)
            df = pd.read_stata(data_path, convert_categoricals=True)[vars_list]

        if verbose:
            print(
                f"Data shape (Original): {df_full.shape[0]} rows x {df_full.shape[1]} columns"
            )
            print(f"Data shape (Cleaned): {df.shape[0]} rows x {df.shape[1]} columns")
            print("-----------------------------------------------")

        dfs[file.stem] = df

```

```python
    return dfs


if __name__ == "__main__":
    for json_name, path_str in data_files.items():
        path = Path(path_str)
        wave_name = json_name.replace(".json", "")
        if path_str.endswith(".sav"):
            # SPSS
            df, meta = pyreadstat.read_sav(
                path, encoding="latin1", apply_value_formats=True
            )

        else:  # Stata .dta file
            df, meta = pyreadstat.read_dta(path, encoding="latin1")

        pd.DataFrame(
            {"variable": meta.column_names, "label": meta.column_labels}
        ).to_csv(Path("output") / f"{wave_name}_variable_descriptions.csv", index=False)

        print(f"--------------- wave {path} ---------------")
        print(df.shape)
        print(df.columns[:20])
        print(
            "Variable/Feature descriptions, csv file saved successfully -> dir=output/"
        )
        df.to_csv((Path("output") / f"{wave_name}.csv"), index=False)

    dfs = load_df_with_var_of_interest(verbose=True)
```

Appendix B

Scenario Cleaning Pipeline Tables

Table B.1: Initial Raw Feature Counts Categorised by Nine Scenarios (Pre Cleaning)

Scenario	**Wave 1**	**Wave 2**	**Wave 3**	**Wave 4**	**Wave 5**	**Harmonised**
Empty / blank values	9	0	7	0	9	0
Numeric with sentinel	187	59	143	60	100	141
Clean numeric	14	27	18	22	20	8
Censored / bracketed numeric	84	71	91	85	85	73
Binary categorical	248	159	177	220	207	187
Multi class categorical / mixed	267	166	203	205	156	44
Single value categorical	0	0	0	0	0	0
Special symbol / mixed	0	0	0	0	0	0
Unknown / complex mixed	0	0	0	0	0	0
Total Columns	**809**	**482**	**639**	**592**	**577**	**453**

Table B.2: Scenario Count Progression: Post Step 1 (Empty / Blank Handling)

Scenario	**Wave 1**	**Wave 2**	**Wave 3**	**Wave 4**	**Wave 5**	**Harmonised**
Empty / blank values	0	0	0	0	0	0
Numeric with sentinel	81	31	42	32	75	15
Clean numeric	124	55	123	50	54	134
Censored / bracketed numeric	84	71	91	85	85	73
Binary categorical	251	159	179	220	207	187
Multi class categorical / mixed	269	166	204	205	156	44
Single value categorical	0	0	0	0	0	0
Special symbol / mixed	0	0	0	0	0	0
Unknown / complex mixed	0	0	0	0	0	0
Total	**809**	**482**	**639**	**592**	**577**	**453**

Table B.3: Scenario Count Progression: Post Step 2 (Sentinel Standardisation)

Scenario	**Wave 1**	**Wave 2**	**Wave 3**	**Wave 4**	**Wave 5**	**Harmonised**
Empty / blank values	0	0	0	0	0	0
Numeric with sentinel	0	0	0	0	0	0
Clean numeric	205	86	165	82	129	149
Censored / bracketed numeric	84	71	91	85	85	73
Binary categorical	251	159	179	220	207	187
Multi class categorical / mixed	269	166	204	205	156	44
Single value categorical	0	0	0	0	0	0
Special symbol / mixed	0	0	0	0	0	0
Unknown / complex mixed	0	0	0	0	0	0
Total	**809**	**482**	**639**	**592**	**577**	**453**

Table B.4: Scenario Count Progression: Post Step 3 (Numeric Cleaning)

Scenario	**Wave 1**	**Wave 2**	**Wave 3**	**Wave 4**	**Wave 5**	**Harmonised**
Empty / blank values	0	0	0	0	0	0
Numeric with sentinel	0	0	0	0	0	0
Clean numeric	263	117	215	124	172	149
Censored / bracketed numeric	16	23	17	16	17	73
Binary categorical	251	160	180	221	207	187
Multi class categorical / mixed	279	182	227	231	181	44
Single value categorical	0	0	0	0	0	0
Special symbol / mixed	0	0	0	0	0	0
Unknown / complex mixed	0	0	0	0	0	0
Total	**809**	**482**	**639**	**592**	**577**	**453**

Table B.5: Scenario Count Progression: Post Step 4 (Censored / Ordinal Mapping)

Scenario	Wave 1	Wave 2	Wave 3	Wave 4	Wave 5	Harmonised
Empty / blank values	0	0	0	0	0	0
Numeric with sentinel	0	0	0	0	0	0
Clean numeric	279	140	233	140	189	222
Censored / bracketed numeric	0	0	0	0	0	0
Binary categorical	251	160	180	221	207	187
Multi class categorical / mixed	279	182	226	231	181	44
Single value categorical	0	0	0	0	0	0
Special symbol / mixed	0	0	0	0	0	0
Unknown / complex mixed	0	0	0	0	0	0
Total	**809**	**482**	**639**	**592**	**577**	**453**

Table B.6: Scenario Count Progression: Post Step 5 (Binary Classification)

Scenario	Wave 1	Wave 2	Wave 3	Wave 4	Wave 5	Harmonised
Empty / blank values	0	0	0	0	0	0
Numeric with sentinel	0	0	0	0	0	0
Clean numeric	530	300	413	361	396	409
Censored / bracketed numeric	0	0	0	0	0	0
Binary categorical	0	0	0	0	0	0
Multi class categorical / mixed	279	182	226	231	181	44
Single value categorical	0	0	0	0	0	0
Special symbol / mixed	0	0	0	0	0	0
Unknown / complex mixed	0	0	0	0	0	0
Total	**809**	**482**	**639**	**592**	**577**	**453**

Table B.7: Final Dfs Cleaned: Post Step 6 (Multi-Class / Mixed Parsing)

Scenario	Wave 1	Wave 2	Wave 3	Wave 4	Wave 5	Harmonised
Empty / blank values	0	0	0	0	0	0
Numeric with sentinel	0	0	0	0	0	0
Clean numeric	809	482	639	592	577	453
Censored / bracketed numeric	0	0	0	0	0	0
Binary categorical	0	0	0	0	0	0
Multi class categorical / mixed	0	0	0	0	0	0
Single value categorical	0	0	0	0	0	0
Special symbol / mixed	0	0	0	0	0	0
Unknown / complex mixed	0	0	0	0	0	0
Total	**809**	**482**	**639**	**592**	**577**	**453**

Appendix C

First Stage: Scenario Variable Mapping

```
# Sentinel values (numeric and string)
# -1 and -1.0 is sentinel but will be used in the clean pipeline
standardised_sentinels = {'-1', '-1.0', -1, -1.0} #  set -> duplicates removed -> counts correct -> single value detection is zero
numeric_sentinels = [-1, -1.0, -98, -98.0, -99, -99.0,
                     95, 95.0, 96, 96.0, 98, 98.0, 99, 99.0]
string_sentinels = [str(x) for x in numeric_sentinels] + [f"_{x}" for x in numeric_sentinels] + \
    ['na', 'n/a', 'refused', 'missing', 'dk', 'rf', 'refused', "refuse", 'none', 'no response', 'nan', 'not answered', 'no answer',
     'not applicable','no consent','dont know',"don't know","didn't answer",'prefer not to say', 'task not done', 'insufficient sample', 'unable to record',
     '__', '*']
blanks = ['', ' ', '\t', '\n']

SCENARIOS = [
    "Empty / blank values",
    "Numeric with sentinel",
    "Clean numeric",
    "Censored / bracketed numeric",
    "Binary categorical",
```

```
    "Multi class categorical / mixed",
    "Single value categorical",
    "Special symbol / mixed",
    "Unknown / complex mixed"
]


# flatten all dtypes across the waves to float and category
def flatten_column_types(df):
    df_flat = df.copy()

    for col in df_flat.columns:
        if pd.api.types.is_numeric_dtype(df_flat[col]): # Numeric columns
            df_flat[col] = df_flat[col].astype(float)  # Convert to float
        else:
            df_flat[col] = (  # Convert to string, strip whitespace, lowercase, then category
                df_flat[col]
                .astype(str)
                .str.strip()
                .str.lower()
                .astype('category')
            )

    return df_flat

def detect_column_scenario_mapping(df, sample_size=10):
    summary = []

    # Regex patterns
    special_symbols_pattern = r'[^A-Za-z0-9\s\.,\-+&/()]' # special symbols excluding common numeric formatting
    bracketed_numeric_pattern = r"\d{1,3}(?:,\d{3})*[-/]\d{1,3}(?:,\d{3})*|\d+\s*[<>+-]"  # patterns like "10-20", "30/40", ">=50"
    sentinel_set = set(x.lower() for x in string_sentinels) # Lowercase string sentinels for comparison

    for col in df.columns:
```

```
        col_data = df[col].copy()
        dtype = str(col_data.dtype) # get data type as string

        # Convert to string for sentinel / blank detection
        str_vals = col_data.astype(str).str.lower().str.strip() # lowercase and strip whitespace
        clean_vals = col_data[~str_vals.isin(sentinel_set) & ~(str_vals.isin(blanks))] # remove sentinels and blanks

        # Remove sentinels and blanks for scenario detection only
        vals_no_sentinel = clean_vals[~clean_vals.astype(str).str.lower().isin(sentinel_set)
                                      & ~clean_vals.astype(str).str.lower().isin(blanks)].unique()
        num_unique_no_sentinel = len(vals_no_sentinel)

        # Check if any sentinel values are present
        sentinel_present = str_vals.isin(sentinel_set).any()

        # Detection flags
        empty_detected = str_vals.isin(blanks).any() or len(col_data) == 0

        # Numeric detection: coerce to numeric, NaNs for non-numeric
        numeric_vals = pd.to_numeric(clean_vals, errors='coerce') # coerce non-numeric to NaN
        numeric_like = numeric_vals.notna().all() and len(clean_vals) > 0 # all values are numeric-like after cleaning

        sentinel_detected = str_vals[~str_vals.isin(standardised_sentinels)].isin(sentinel_set).any() # or col_data.eq(-1.0).any() # check for sentinel presence
        censored_detected = clean_vals.astype(str).str.contains(bracketed_numeric_pattern).any() # check for bracketed numeric patterns

        # Binary categorical detection (exactly 2 unique values after cleaning)
```

```
        binary_categorical_detected = False
        if dtype == 'category':
            if num_unique_no_sentinel == 2:
                binary_categorical_detected = True
            elif num_unique_no_sentinel == 1 and sentinel_present:
                binary_categorical_detected = True

        # Multi-class categorical (3+ unique values, object/category)
        multi_class_categorical_detected = num_unique_no_sentinel > 2 and dtype == 'category'

        # Single value categorical
        single_value_detected = num_unique_no_sentinel == 1 and not binary_categorical_detected

        # Special symbols detection (ignore numeric formatting)
        symbol_detected = clean_vals.astype(str).str.contains(special_symbols_pattern).any()

        # Mixed types detection
        mixed_detected = len(set(type(v) for v in col_data)) > 1  # more than one data type present

      # Determine scenario in order of pipeline
        if empty_detected:
            scenario = SCENARIOS[0]
            handling = "Replace numeric/categorical empty/blank with -1.0"
        elif numeric_like and sentinel_detected:
            scenario = SCENARIOS[1]
            handling = "Replace numeric sentinel values with -1.0"
        elif numeric_like and not sentinel_detected:
            scenario = SCENARIOS[2]
            handling = "Keep as numeric"
        elif censored_detected:
            scenario = SCENARIOS[3]
            handling = "Convert numeric ranges to midpoints or keep as categorical"
```

```
        elif binary_categorical_detected:
            scenario = SCENARIOS[4]
            handling = "Encode as 0/1 or keep as category"
        elif multi_class_categorical_detected:
            scenario = SCENARIOS[5]
            handling = "One-hot encode, label encode, or keep as category"
        elif single_value_detected:
            scenario = SCENARIOS[6]
            handling = "Keep as category / constant column"
        elif symbol_detected:
            scenario = SCENARIOS[7]
            handling = "Inspect and preprocess based on context"
        else:
            scenario = SCENARIOS[8]
            handling = "Inspect manually / complex mixed types"

        sample_values = list(col_data.unique()[:sample_size]) # sample unique values

        summary.append({
            'Variable': col,
            'DataType': dtype,
            'UniqueValues': num_unique_no_sentinel,
            'Scenario': scenario,
            'SuggestedHandling': handling,
            'SampleValues': sample_values
        })

    return pd.DataFrame(summary)
```

Appendix D

Second Stage: Deterministic six step cleaning engine

```
# Cleaning pipeline to all waves
dfs_cleaned = {}

# flattened dtype df
dfs_flat = {wave: flatten_column_types(df) for wave, df in dfs.items()} # flatten dtypes

# Original df
display(scenario_count_table(dfs_flat).style.set_caption("Original Table - pre cleaning")) # display original scenario count table

# Step 1: Clean empty/missing values
dfs_step1 = {wave: clean_empty_missing(df) for wave, df in dfs_flat.items()} # apply step 1 cleaning
display(scenario_count_table(dfs_step1).style.hide(axis="index").set_caption("Step 1: Empty / Blank - post cleaning"))

# Step 2: Clean sentinel values
dfs_step2 = {wave: clean_sentinel(df) for wave, df in dfs_step1.items()} # apply step 2 cleaning
display(scenario_count_table(dfs_step2).style.hide(axis="index").set_caption("Step 2: Sentinel - post cleaning"))

```

```python
# Step 3: Numerical Transformation
dfs_step3 = {wave: numerical_transform(df) for wave, df in dfs_step2.items()} # apply step 3 cleaning
display(scenario_count_table(dfs_step3).style.hide(axis="index").set_caption("Step 3: Numeric Transformation - post cleaning"))

# Step 4: Categorical Transformation for censored scenario
dfs_step4 = {wave: categorical_transform_S4(df) for wave, df in dfs_step3.items()} # apply step 4 cleaning
display(scenario_count_table(dfs_step4).style.hide(axis="index").set_caption("Step 4: Category Transformation S4 - post cleaning"))

# Step 5: Categorical Transformation for binary values
dfs_step5 = {wave: categorical_transform_S5(df) for wave, df in dfs_step4.items()} # apply step 5 cleaning
display(scenario_count_table(dfs_step5).style.hide(axis="index").set_caption("Step 5: Category Transformation S5 - post cleaning"))

# Step 6: Categorical Transformation for multi class/mixed values
dfs_step6 = {wave: categorical_transform_S6(df) for wave, df in dfs_step5.items()} # apply step 6 cleaning
display(scenario_count_table(dfs_step6).style.hide(axis="index").set_caption("Step 6: Category Transformation S6 - post cleaning"))

dfs_cleaned = dfs_step6


def show_grouped_vars_for_scenarios(dfs, scenarios_to_show): # Function to show grouped variables per scenario across waves -> markdown format
    """
    Used to inspect the scenario variables/features that need handling for transformation/cleaning.
    A pre and post cleaning helper function to identify variables that need specific handling and to
    Inspect actual changes for a scenario from previous cleaning step to current step.
    """
```

```python

    print("# Category Transformation Mapping")
    print(f"## Scenario: {scenarios_to_show}") # Markdown mapping dictionary header
    for wave, df in dfs.items():
        print(f"\n### =================== {wave.upper()} ====================")

        mapping_df = detect_column_scenario_mapping(df) # Get scenario mapping

        for scenario in scenarios_to_show:
            vars_in_scenario = mapping_df[mapping_df['Scenario'] == scenario]['Variable'].tolist() # Variables in scenario
            # print(f"\n--- Scenario: {scenario} ---") # Scenario header for pipeline git commit
            print(f"\n--- Total: {len(vars_in_scenario)} ---") # \n
            if not vars_in_scenario:
                print("No variables found.")
                continue

            # Dictionary: frozen set(values) -> list of variables
            groups = {}

            for var in vars_in_scenario:
                unique_vals = list(dict.fromkeys(df[var].dropna()))  # Preserve first-seen order
                key = tuple(unique_vals)  # Use ordered tuple as key

                if key not in groups: # Initialise list if key not present
                    groups[key] = []
                groups[key].append(var) # Append variable to the group

            # Print grouped variables in markdown format friendly for Notion -> mapping dictionary
            for value_set, var_list in groups.items():
```

```
                print("\n*Variables:*","`" + ", ".join(var_list) + "`  ") # \n
                print(f"*Pre transformed values:* {list(value_set)}  ")

                # Pull transformed values from dfs_cleaned / specific step using preserved order
                try:
                    df_before = df[var_list[0]]
                    df_after = dfs_cleaned[wave][var_list[0]]

                    # Build transformed list in same order as original values
                    transformed_vals = []
                    seen = set()

                    for orig_val in value_set: # Iterate through original values in order
                        mask = df_before == orig_val
                        if mask.any():
                            trans_vals = df_after[mask].unique().tolist() # Get unique transformed values
                            if trans_vals:
                                transformed_vals.append(trans_vals[0]) # Append first transformed value
                except Exception:
                    transformed_vals = []
                print(f"*Transformed numeric:* `{transformed_vals}`")


scenarios_to_show = SCENARIOS
# scenarios_to_show = ["Clean numeric"]
# scenarios_to_show = ["Censored / bracketed numeric"]
# scenarios_to_show = ["Binary categorical"]
# scenarios_to_show = ["Multi class categorical / mixed"]

show_grouped_vars_for_scenarios(dfs_cleaned, scenarios_to_show) # prev step == comparison to current step else current step
```

```
    transformed/cleaned for ready .md dictionary
```

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